

# Tutorial

## Software Reuse

1. Suggest why the savings in cost from reusing existing software are not simply proportional to the size of the components that are reused.
2. What are the differences between SPL and COTS?
3. Testing process for COTS-solution system should focus on the interactions between the operational processes and the system configuration. Justify.
4. Imagine you're building a dynamic web app—say, a live music playlist editor—using a framework like React or Django. Your app needs to respond to user actions and lifecycle events. Dive into the roles of callbacks and hooks: how do they work in a web framework, and what sets them apart in bringing your app to life?
5. A software development team is tasked with creating a fitness tracking application that supports multiple user roles (athletes, coaches, health professionals), includes workout planning features, and integrates with external wearable devices (e.g., Fitbit). The client demands delivery within five months, prioritizes maintainability, and possible features extension in the future.  
Complete the following structured short-answer questions based on the scenario:
  - Name **one** software reuse technique suitable for this project.
  - State **two reasons** why the technique you chose in part 1) is suitable for this project.
6. You are a software engineering consultant hired by a company that develops mobile applications for different industries, such as healthcare, finance, and education. The company is considering adopting software reuse techniques to reduce development time and costs while maintaining high quality. They are particularly interested in understanding the differences between Application Frameworks and Software Product Lines (SPL).

Scenario:

The company plans to develop a new set of mobile apps for small businesses, including:

- A point-of-sale (POS) app for retail stores.
- An appointment scheduling app for service-based businesses.
- An inventory management app for warehouses.

All apps need to run on both iOS and Android, and they must share common features like user authentication, payment processing, and data analytics.

Tasks:

- Define Application Frameworks and Software Product Lines (SPL), highlighting their key differences.
- Recommend which technique (Application Frameworks or SPL) is more suitable for the company's needs. Justify your choice with at least three specific reasons based on the scenario.
- Identify two potential challenges the company might face when implementing your recommended technique.

## Additional Questions

### Question 1

A software development team is building a travel booking platform that supports multiple user roles (travelers, agents, administrators), includes trip planning features, and integrates with external APIs (e.g., flight data from Amadeus). The client requires delivery within six months and prioritizes ease of updates to adapt to new travel regulations.

#### Task:

Complete the following structured short-answer questions based on the scenario:

1. Name **one** software reuse technique suitable for this project.
2. State **two reasons** why the technique you chose in part 1) is suitable for this project.

### Question 2

A software development team is developing an event management system that supports multiple user roles (organizers, attendees, vendors), includes ticketing features, and integrates with external payment gateways (e.g., Stripe). The client insists on delivery within three months and prioritizes rapid deployment to capture an upcoming event season.

#### Task:

Complete the following structured short-answer questions based on the scenario:

1. Name **one** software reuse technique suitable for this project.
2. State **two reasons** why the technique you chose in part 1) is suitable for this project.

### Question 3

A software development team is creating a suite of smart home applications supporting multiple user roles (residents, installers, support staff), including features for device control (e.g., lights, thermostats), and integrating with external IoT systems (e.g., Amazon Alexa). The client demands delivery within eight months and prioritizes consistency across the suite for branding purposes.

#### Task:

Complete the following structured short-answer questions based on the scenario:

1. Name **one** software reuse technique suitable for this project.
2. State **two reasons** why the technique you chose in part 1) is suitable for this project.